

CSCI3340 Scientific Computing

2025–26 Term 2

Assignment 3

Submission Deadline: 23:59:00, 16th April 2026.

Submission Guidelines

Assignment 3 covers material from Lectures 7-10 and is worth a total of 100 points. There are 6 problems in total: ALL of them are programming tasks.

You should submit a single .zip file containing exactly one PDF file and all relevant MATLAB files. The PDF must include your solutions to all problems and should be self-contained, clear, and easy to read. You may write your answers by hand or type them. For each programming problem, clearly indicate in your PDF which MATLAB file corresponds to that problem. Your PDF should also include brief instructions on how to run your code and interpret the results. If not specified otherwise, you are allowed to use any built-in functions.

If the submission was late but within 3 days after the deadline, a -30 points penalty will be applied. Submissions made after this period will not be graded.

If you have any concerns, feel free to contact me (colintsang@cuhk.edu.hk) or the TA in charge (srijan25@link.cuhk.edu.hk).

Problem 1 (15 points)

The expected number of uniform random numbers drawn from $[0, 1]$ needed until their sum exceeds 1 is exactly $e \approx 2.71828$. Write a MATLAB script that estimates e using $T = 500,000$ trials. For each trial, accumulate `rand()` values until their sum exceeds 1 and record the count N . Report:

- (a) The estimated value of e (sample mean $\hat{\mu}$).
- (b) The absolute error against `exp(1)`.
- (c) The 95% confidence interval: $\hat{\mu} \pm \frac{1.96 \hat{\sigma}}{\sqrt{T}}$.

Problem 2 (20 points)

Model a stock price using geometric Brownian motion. Starting at price 100, at each daily step the log-return is drawn from $N(\mu, \sigma^2)$ where $\mu = 0.0003$ and $\sigma = 0.01$. Write a MATLAB script that simulates 1,000 price paths over 252 trading days (1 year). Plot all

paths with low opacity, and overlay the mean path. Report the 5th and 95th percentile of final prices as a shaded region.

NOTE: You are not allowed to use the built-in function `gbm()`.

Problem 3 (15 points)

A gambler starts with 50 and plays a game where they win 1 with probability $p = 0.48$ and lose 1 with probability 0.52 each round. They stop when they either reach 100 or go broke. Write a MATLAB script to simulate 10,000 games and estimate:

- (a) The probability of reaching 100.
- (b) The average number of rounds played before stopping.

Problem 4 (15 points)

Solve the equation

$$f(x) = \int_0^1 e^{xt^2} dt - 2 = 0$$

using Newton's method. Compute f and f' with Adaptive Quadrature Using Simpson's Rule.

NOTE: You are not allowed to use the built-in functions `integral()`, `quad()`, `quadl()`, `fzero()`, `trapz()`, `fsolve()`, `root()`, `solve()`.

Problem 5 (20 points)

In a chemical reaction, one molecule of A combines with one molecule of B to form one molecule of the chemical C. It is found that the concentration $y(t)$ of C at time t is the solution to the I.V.P.

$$y' = k(a - y)(b - y) \quad \text{with} \quad y(0) = 0,$$

where k is a positive constant and a and b are the initial concentrations of A and B, respectively. Suppose that $k = 0.01$, $a = 70$ millimoles/liter, and $b = 50$ millimoles/liter. Use the Runge-Kutta method of order $N = 4$ with $h = 0.5$ to find the solution over $[0, 20]$. *Remark.* You can compare your computer solution with the exact solution $y(t) = 350(1 - e^{-0.2t})/(7 - 5e^{-0.2t})$. Observe that the limiting value is 50 as $t \rightarrow +\infty$.

NOTE: You are not allowed to use any built-in ODE solvers like `ode45()`, `ode23()`, `dsolve()`, ..., etc.

Problem 6 (15 points)

The Poisson equation can be written in three dimensions as

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = f(x, y, z)$$

Solve for the distribution of temperature within a unit (1×1) cube with zero boundary conditions and $f = -10$. Employ

$$\Delta x = \Delta y = \Delta z = \frac{1}{6}.$$

and by implementing Finite-Difference Method in MATLAB.

NOTE: You are not allowed to use the entire PDE toolbox.

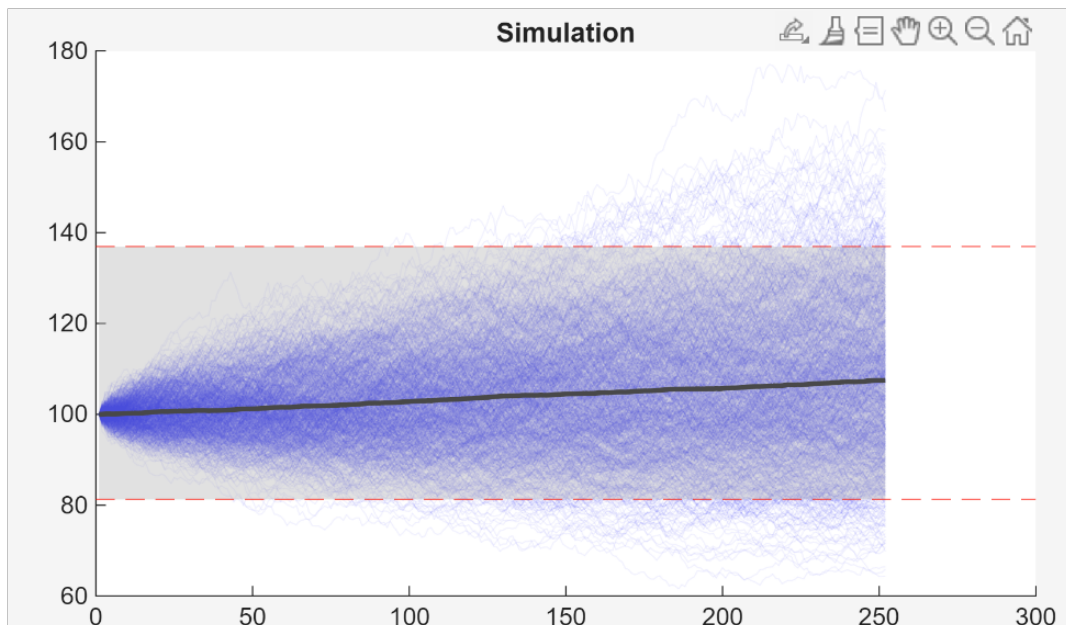
Problem 1:

The script is in q1.m and the output result is shown below:

```
>> q1
(1) Estimate: 2.71843
(2) Absolute Error: 0.00014
(3) 95% CI: [2.71600, 2.72085]
```

Problem 2:

The script is in q2.m and the output result is shown below (black line is mean, red line is the 5% and 95% percentile):



```
>> q2
5% percentile: 81.36467, 95% percentile: 139.12727
```

Problem 3:

The script is in q3.m and the output result is shown below:

- (a) Probability of reaching 100: 0.0178
- (b) Average rounds before stopping: 1191.04

Problem 4:

The script is in q4.m and the output result is shown below. Initial value is $x=1$. It took 5 iterations for the Newton's Method, the intermediate f and f' value, and the final solution is shown below:

At iteration 1:

$x = 1.000000$

$f = -0.537348$

$f' = 0.627815$

At iteration 2:

$x = 1.855902$

$f = 0.192496$

$f' = 1.132864$

At iteration 3:

$x = 1.685982$

$f = 0.011162$

$f' = 1.004337$

At iteration 4:

$x = 1.674869$

$f = 0.000044$

$f' = 0.996511$

At iteration 5:

$x = 1.674825$

$f = 0.000000$

$f' = 0.996480$

Root by Newton Method: 1.674825

Problem 5:

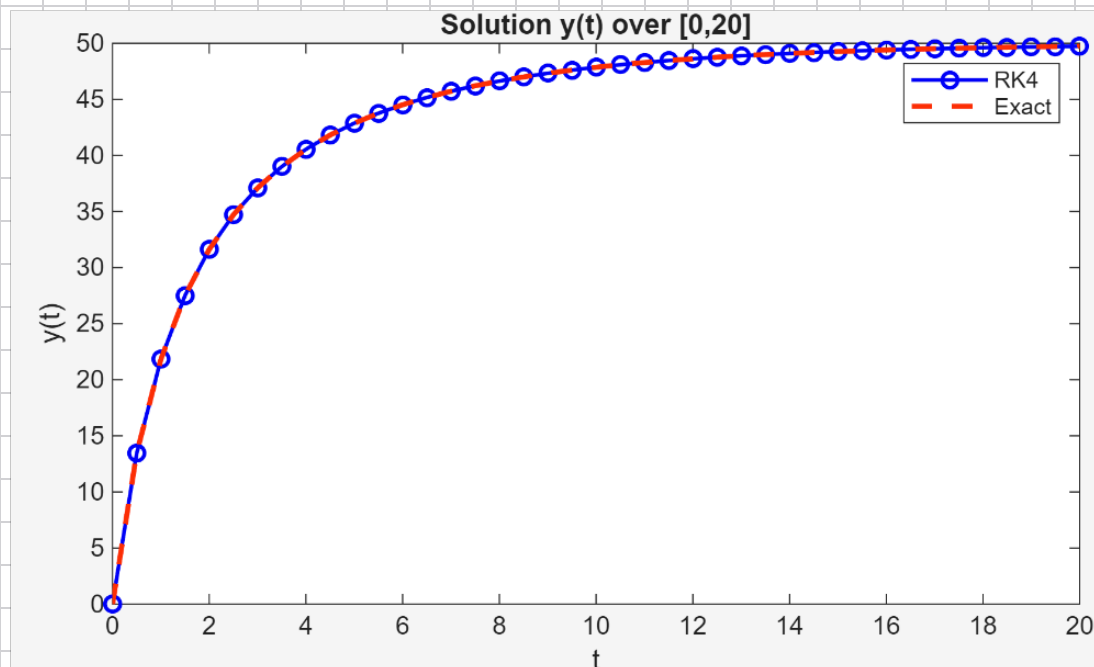
The script is in q5.m and the output result is shown below.

The solution is:

t	RK4 y	Exact y	Abs Error
0.0	0.000000	0.000000	0.00e+00
0.5	13.451086	13.452916	1.83e-03
1.0	21.827765	21.829552	1.79e-03
1.5	27.521617	27.523098	1.48e-03
2.0	31.625820	31.627012	1.19e-03
2.5	34.710971	34.711932	9.61e-04
3.0	37.104031	37.104813	7.83e-04
3.5	39.005837	39.006482	6.45e-04
4.0	40.546583	40.547120	5.37e-04
4.5	41.814382	41.814833	4.51e-04
5.0	42.871004	42.871386	3.83e-04
5.5	43.761044	43.761371	3.27e-04
6.0	44.517508	44.517790	2.81e-04
6.5	45.165353	45.165597	2.44e-04
7.0	45.723799	45.724010	2.12e-04
7.5	46.207889	46.208074	1.85e-04
8.0	46.629569	46.629732	1.63e-04
8.5	46.998443	46.998586	1.43e-04
9.0	47.322320	47.322447	1.26e-04

t	RK4 y	Exact y	Abs Error
9.5	47.607616	47.607728	1.12e-04
10.0	47.859645	47.859744	9.94e-05
10.5	48.082851	48.082939	8.85e-05
11.0	48.280972	48.281051	7.89e-05
11.5	48.457179	48.457249	7.05e-05
12.0	48.614171	48.614234	6.30e-05
12.5	48.754264	48.754320	5.64e-05
13.0	48.879453	48.879504	5.06e-05
13.5	48.991464	48.991510	4.54e-05
14.0	49.091797	49.091838	4.08e-05
14.5	49.181760	49.181797	3.67e-05
15.0	49.262497	49.262530	3.31e-05
15.5	49.335013	49.335043	2.98e-05
16.0	49.400193	49.400220	2.68e-05
16.5	49.458816	49.458841	2.42e-05
17.0	49.511575	49.511596	2.18e-05
17.5	49.559079	49.559099	1.97e-05
18.0	49.601873	49.601891	1.78e-05
18.5	49.640441	49.640457	1.61e-05
19.0	49.675213	49.675228	1.45e-05
19.5	49.706574	49.706588	1.31e-05
20.0	49.734868	49.734880	1.19e-05

The comparison with the exact solution (red is exact, blue is calculated):



Problem 6:

The script for solving the equation is in q6.m. The final T is the solution to the equation in the problem. Following the visualization method in Lab 10, I also visualize the solution T when $z=0.5$:

